03_predictive_model.R

Complete source code for reference. The original .R file must remain in GitHub for execution.

```
# Predictive modeling and diagnostic evidence for campaign revenue.
# Base R only: no external packages are required.

data <- read.csv("outputs/clean_advertising_campaigns.csv", stringsAsFactors = FALSE)
data$platform <- factor(data$platform)
data$campaign_type <- factor(data$campaign_type)
data$audience <- factor(data$audience)
train <- subset(data, split == "train")
test <- subset(data, split == "test")

model <- lm(revenue ~ spend + impressions + clicks + conversions + platform + campaign_type + audience, data = train)
pred <- predict(model, newdata = test)

rmse <- sqrt(mean((test$revenue - pred)^2))
mae <- mean(abs(test$revenue - pred))
test_r2 <- 1 - sum((test$revenue - pred)^2) / sum((test$revenue - mean(test$revenue))^2)
sm <- summary(model)
metrics <- data.frame(
  metric = c("Train_R2", "Adjusted_R2", "Test_R2", "RMSE", "MAE", "F_statistic", "F_p_value"),
  value = c(sm$r.squared, sm$adj.r.squared, test_r2, rmse, mae,
            unname(sm$fstatistic[1]), pf(sm$fstatistic[1], sm$fstatistic[2], sm$fstatistic[3], lower.tail = FALSE))
)
write.csv(metrics, "outputs/model_metrics.csv", row.names = FALSE)

coefs <- as.data.frame(sm$coefficients)
coefs$term <- rownames(coefs)
rownames(coefs) <- NULL
names(coefs) <- c("estimate", "std_error", "t_value", "p_value", "term")
coefs <- coefs[, c("term", "estimate", "std_error", "t_value", "p_value")]
write.csv(coefs, "outputs/model_coefficients.csv", row.names = FALSE)
capture.output(sm, file = "outputs/model_summary.txt")

predictions <- data.frame(actual_revenue = test$revenue, predicted_revenue = as.numeric(pred))
write.csv(predictions, "outputs/test_predictions.csv", row.names = FALSE)

png("outputs/plots/actual_vs_predicted_revenue.png", width=1000, height=600)
plot(test$revenue, pred, pch=19, xlab="Actual Revenue", ylab="Predicted Revenue", main="Actual vs Predicted Revenue")
abline(0,1,lwd=2)
dev.off()

png("outputs/plots/diagnostic_residuals_vs_fitted.png", width=1000, height=600)
plot(model, which=1, main="Residuals vs Fitted")
dev.off()

png("outputs/plots/diagnostic_qq_plot.png", width=1000, height=600)
plot(model, which=2, main="Normal Q-Q Plot")
dev.off()

png("outputs/plots/diagnostic_scale_location.png", width=1000, height=600)
plot(model, which=3, main="Scale-Location")
dev.off()

png("outputs/plots/diagnostic_residuals_vs_leverage.png", width=1000, height=600)
plot(model, which=5, main="Residuals vs Leverage")
dev.off()

cat("Predictive model completed. Test RMSE:", round(rmse, 2), "Test MAE:", round(mae, 2), "Test R-squared:", round(test_
```